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File Compressor and Decompressor Using Java

Introduction

File compression is a process of reducing the size of a file to save storage space and improve file transfer efficiency. Compressed files occupy less disk space and can be transmitted faster over networks. Decompression restores the compressed file back to its original format.

This project aims to develop a simple tool that allows users to compress files into ZIP format and extract ZIP files using Java. The system provides a graphical user interface to make file operations easier for users.

Abstract

The File Compressor and Decompressor is a Java-based application that enables users to compress files into ZIP format and extract them when required. The system uses Java's built-in java.util.zip library to perform compression and decompression operations. A simple graphical user interface developed using Java Swing allows users to select files and perform compression or extraction easily. This project demonstrates the use of file handling, data compression techniques, and GUI development in Java.

Tools and Technologies Used

Technology	Purpose
Java	Core programming language
Java Swing	Graphical user interface
java.util.zip	File compression and extraction
Eclipse IDE	Development environment

Steps Involved in Building the Project

1. Created a Java project in Eclipse.

2. Developed a graphical user interface using Java Swing.
3. Implemented file selection using JFileChooser.
4. Used ZipOutputStream to compress selected files into ZIP format.
5. Used ZipInputStream to extract compressed files.
6. Implemented buffer streams to efficiently read and write file data.
7. Tested the application with different file types to verify compression and extraction functionality.

Conclusion

The File Compressor and Decompressor application successfully demonstrates file compression and decompression using Java. The system allows users to easily compress files to reduce storage space and extract ZIP files when needed. The project highlights important programming concepts such as file handling, stream processing, and graphical user interface development. This project enhanced understanding of Java libraries and practical implementation of data compression techniques.